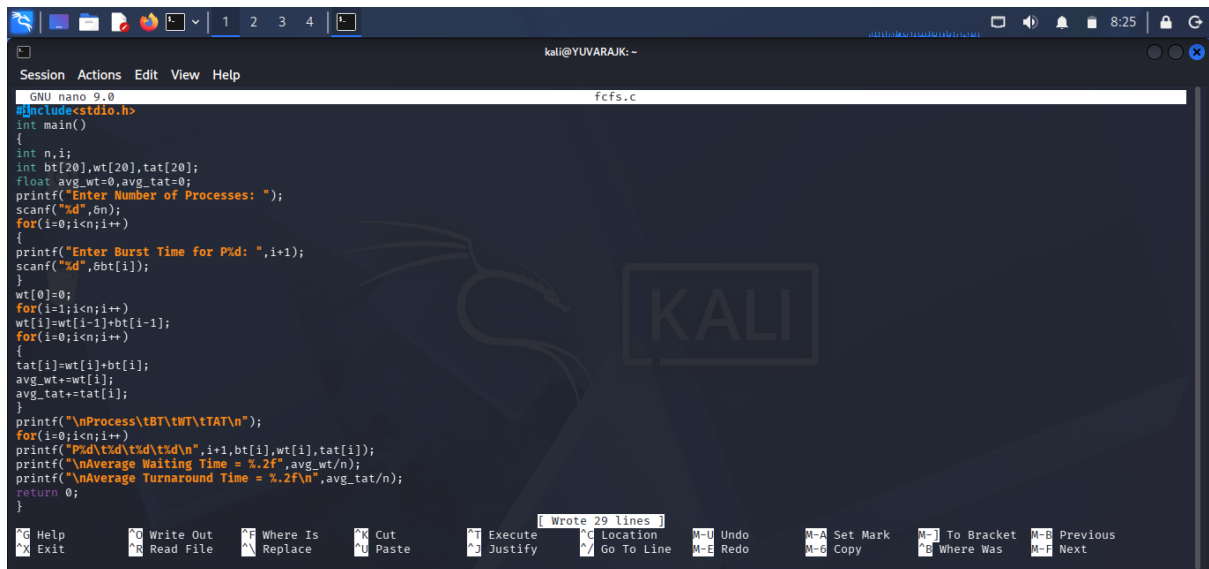
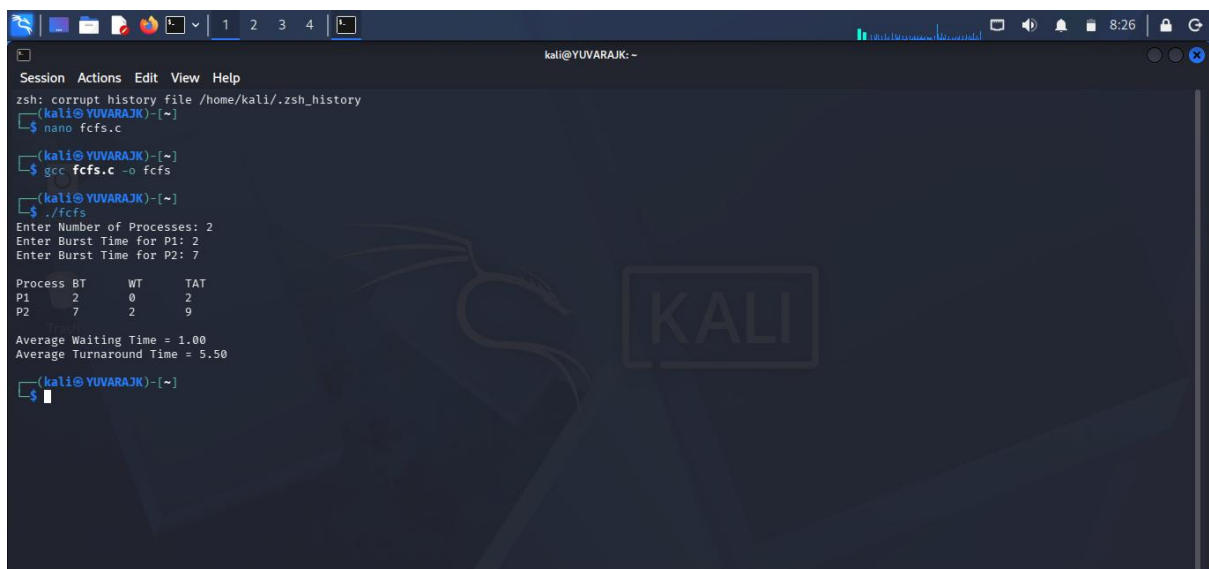


C PROGRAMMING : FCFS



```
GNU nano 9.0 fcfs.c
#include<stdio.h>
int main()
{
    int n,i;
    int bt[20],wt[20],tat[20];
    float avg_wt=0,avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }
    wt[0]=0;
    for(i=1;i<n;i++)
    {
        wt[i]=wt[i-1]+bt[i-1];
    }
    for(i=0;i<n;i++)
    {
        tat[i]=wt[i]+bt[i];
        avg_wt+=wt[i];
        avg_tat+=tat[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
    }
    printf("\nAverage Waiting Time = %.2f",avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n",avg_tat/n);
    return 0;
}
```

OUTPUT :

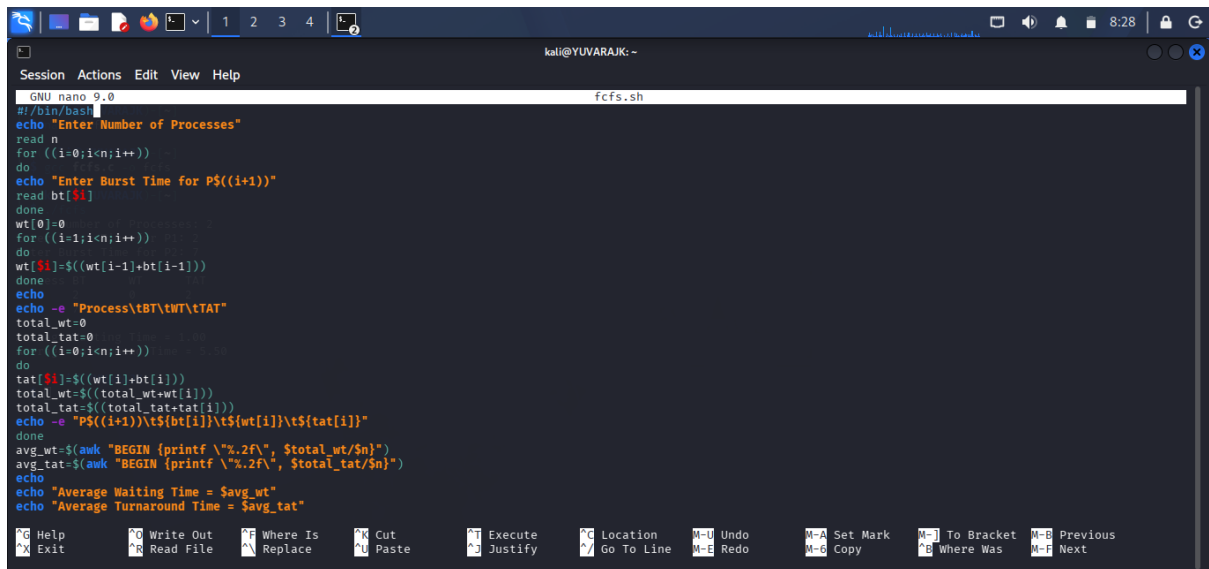


```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)~$ nano fcfs.c
(kali@YUVARAJK)~$ gcc fcfs.c -o fcfs
(kali@YUVARAJK)~$ ./fcfs
Enter Number of Processes: 2
Enter Burst Time for P1: 2
Enter Burst Time for P2: 7

Process BT    WT    TAT
P1      2      0      2
P2      7      2      9

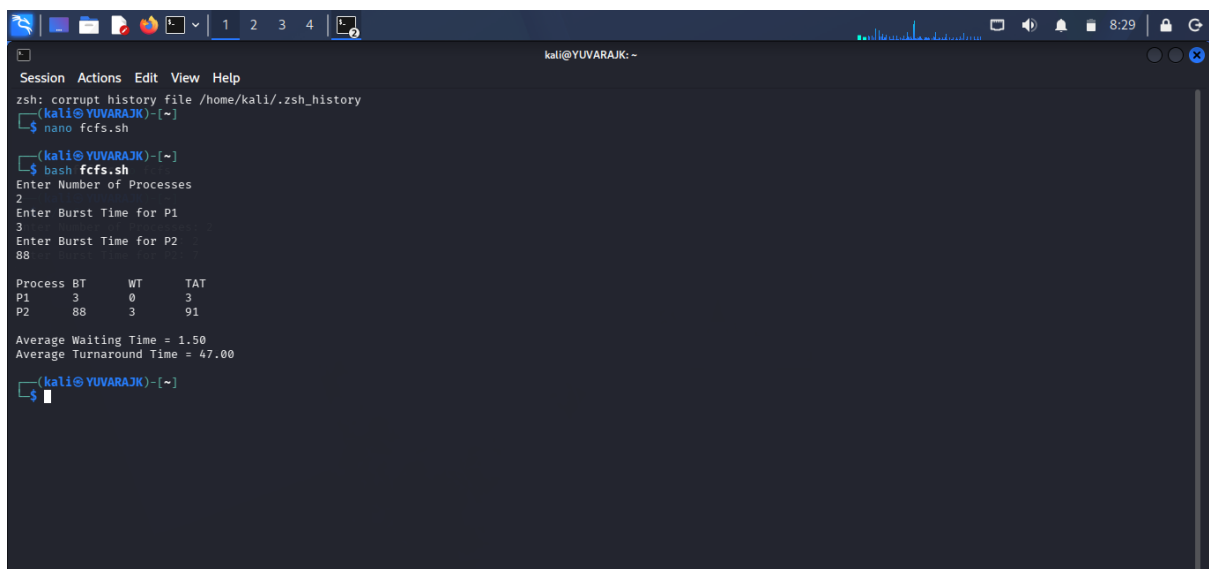
Average Waiting Time = 1.00
Average Turnaround Time = 5.50
(kali@YUVARAJK)~$
```

SHELL PROGRAMMING :FCFS



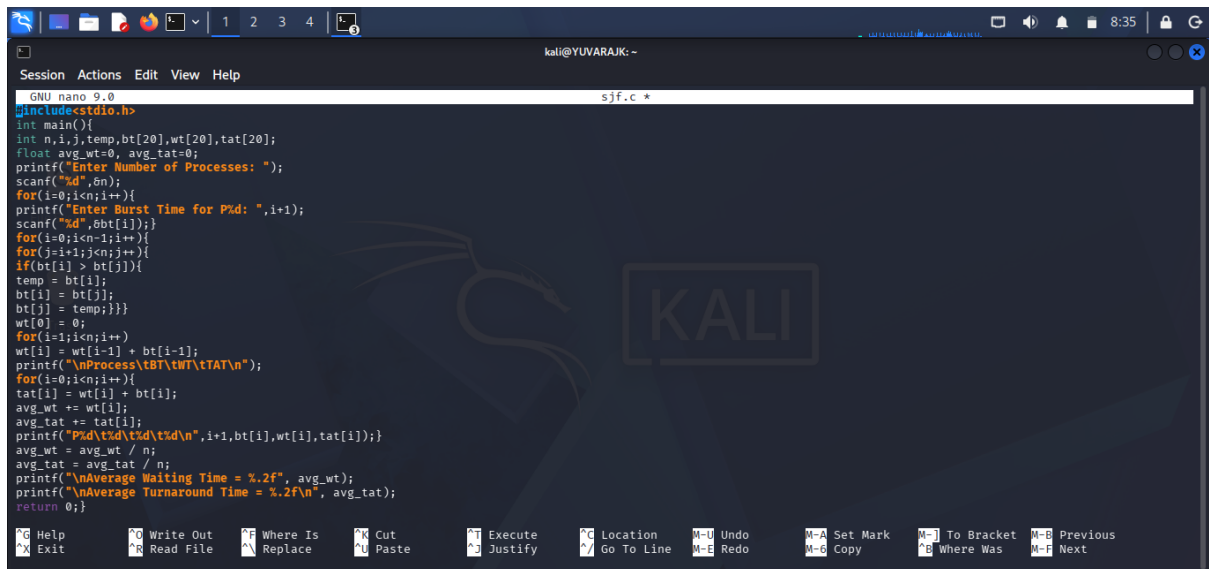
```
GNU nano 9.0 fcfs.sh
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P${(i+1)}"
read bt[i]
done
wt[0]=0
for ((i=1;i<n;i++))
do
wt[i]=$((wt[i-1]+bt[i-1]))
done
echo
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
tat[i]=$((wt[i]+bt[i]))
total_wt=$((total_wt+wt[i]))
total_tat=$((total_tat+tat[i]))
echo -e "P${(i+1)}\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt/$n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat/$n}")
echo
echo "Average Waiting Time = $avg_wt"
echo "Average Turnaround Time = $avg_tat"
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano fcfs.sh
(kali@YUVARAJK)-[~]
$ bash fcfs.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
88
Process BT      WT      TAT
P1      3        0        3
P2      88       3       91
Average Waiting Time = 1.50
Average Turnaround Time = 47.00
(kali@YUVARAJK)-[~]
$
```

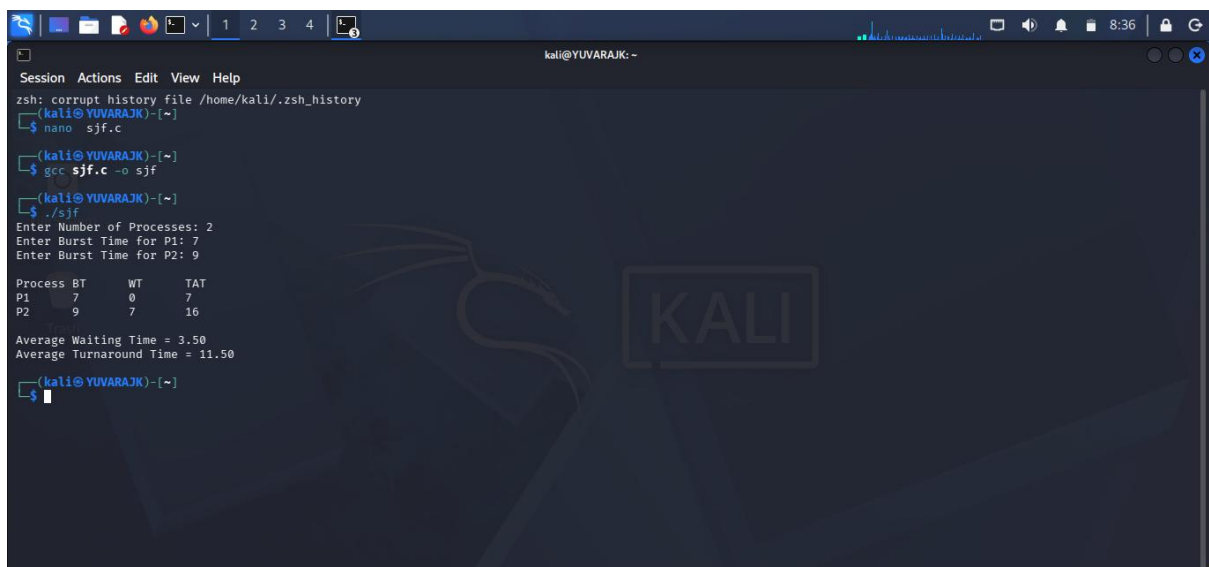
C PROGRAMMING :SJF



The screenshot shows a terminal window with the nano 9.0 editor open, editing a file named sjf.c. The code implements the Shortest Job First (SJF) scheduling algorithm. It takes the number of processes (n) and their burst times (bt) as input. It then calculates the waiting time (wt) and turnaround time (tat) for each process based on the shortest burst time priority. The program outputs the average waiting time and average turnaround time.

```
GNU nano 9.0 sjf.c *
#include<stdio.h>
int main(){
    int n,i,j,temp,bt[20],wt[20],tat[20];
    float avg_wt=0, avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }
    for(i=0;i<n-1;i++){
        for(j=i+1;j<n;j++){
            if(bt[i] > bt[j]){
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
        wt[i] = 0;
        for(i=1;i<n;i++){
            wt[i] = wt[i-1] + bt[i-1];
            printf("\nProcess\tBT\tWT\tTAT\n");
        }
        for(i=0;i<n;i++){
            tat[i] = wt[i] + bt[i];
            avg_wt += wt[i];
            avg_tat += tat[i];
            printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
        }
        avg_wt = avg_wt / n;
        avg_tat = avg_tat / n;
        printf("\nAverage Waiting Time = %.2f", avg_wt);
        printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
        return 0;
    }
```

OUTPUT :



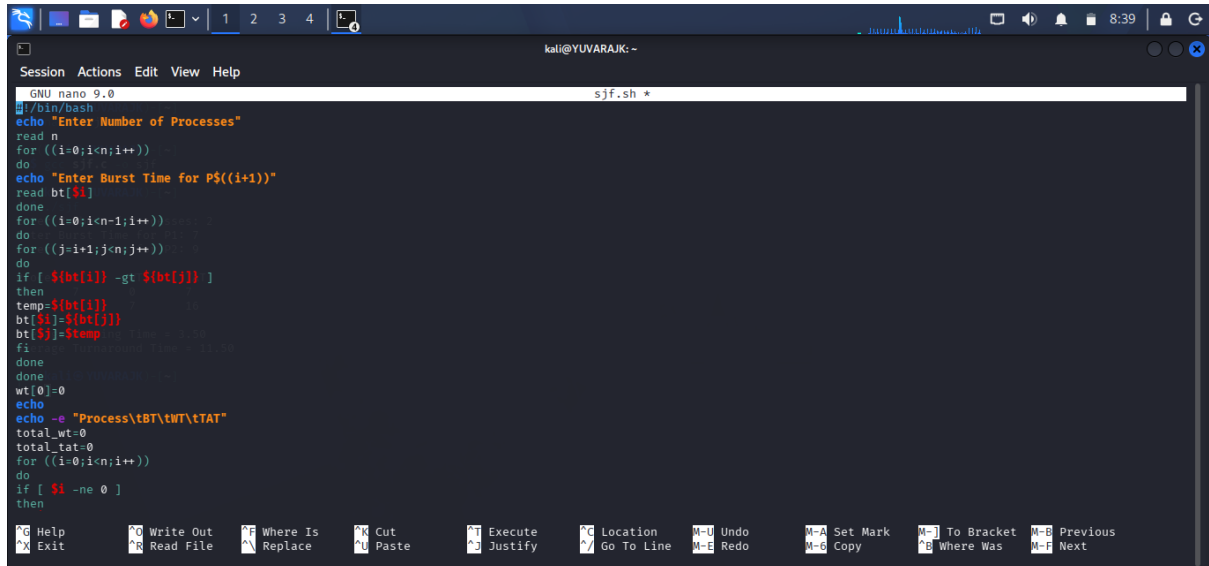
The screenshot shows the terminal output of the SJF program. The user enters 2 for the number of processes. The burst times for P1 and P2 are entered as 7 and 9 respectively. The program then displays the waiting time (WT) and turnaround time (TAT) for each process. The average waiting time is 3.50 and the average turnaround time is 11.50.

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano sjf.c
(kali@YUVARAJK)-[~]
$ gcc sjf.c -o sjf
(kali@YUVARAJK)-[~]
$ ./sjf
Enter Number of Processes: 2
Enter Burst Time for P1: 7
Enter Burst Time for P2: 9

Process BT      WT      TAT
P1      7        0       7
P2      9        7      16

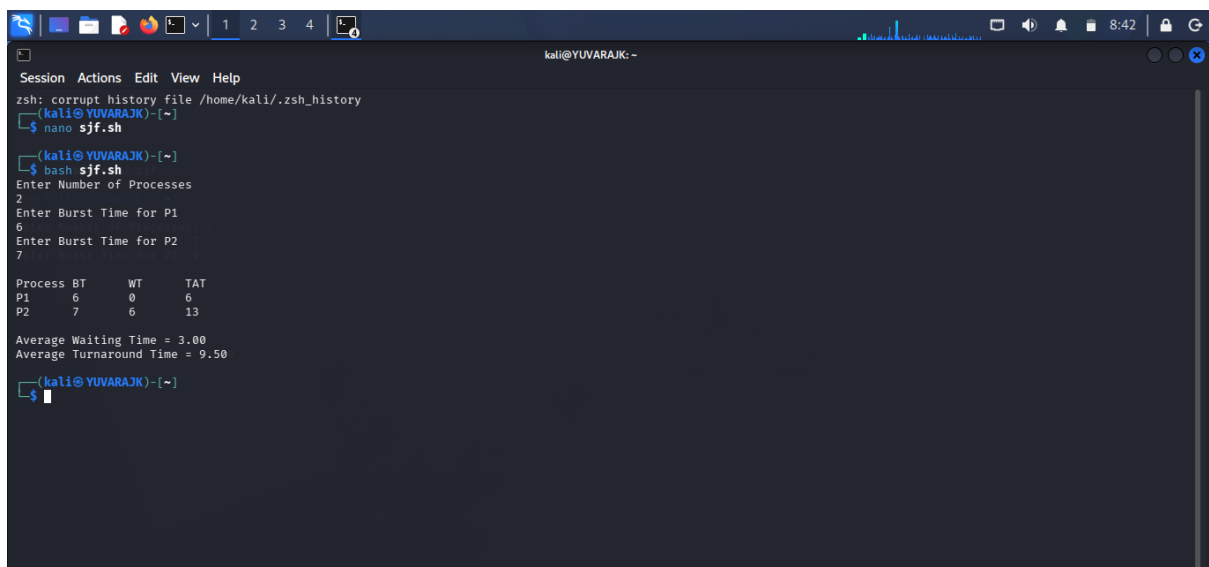
Average Waiting Time = 3.50
Average Turnaround Time = 11.50
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : SJF



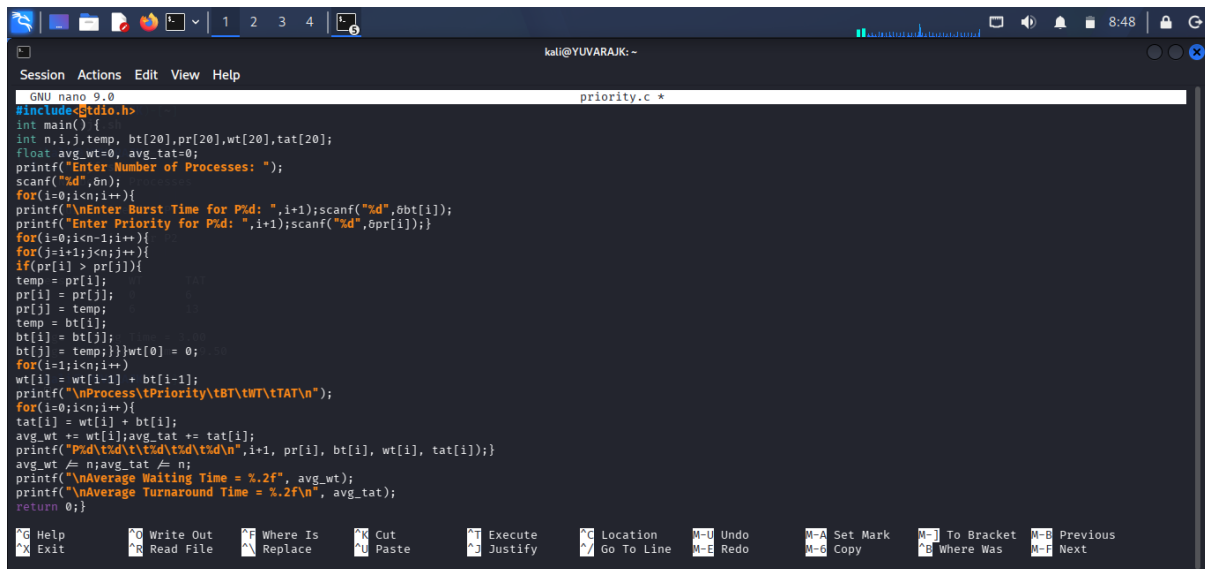
```
GNU nano 9.0 sjf.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${bt[i]} -gt ${bt[j]} ]
then
temp=${bt[i]}
bt[i]=${bt[j]}
bt[j]=$temp
fi
done
done
wt[0]=0
echo
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
if [ $i -ne 0 ]
then
```

OUTPUT :



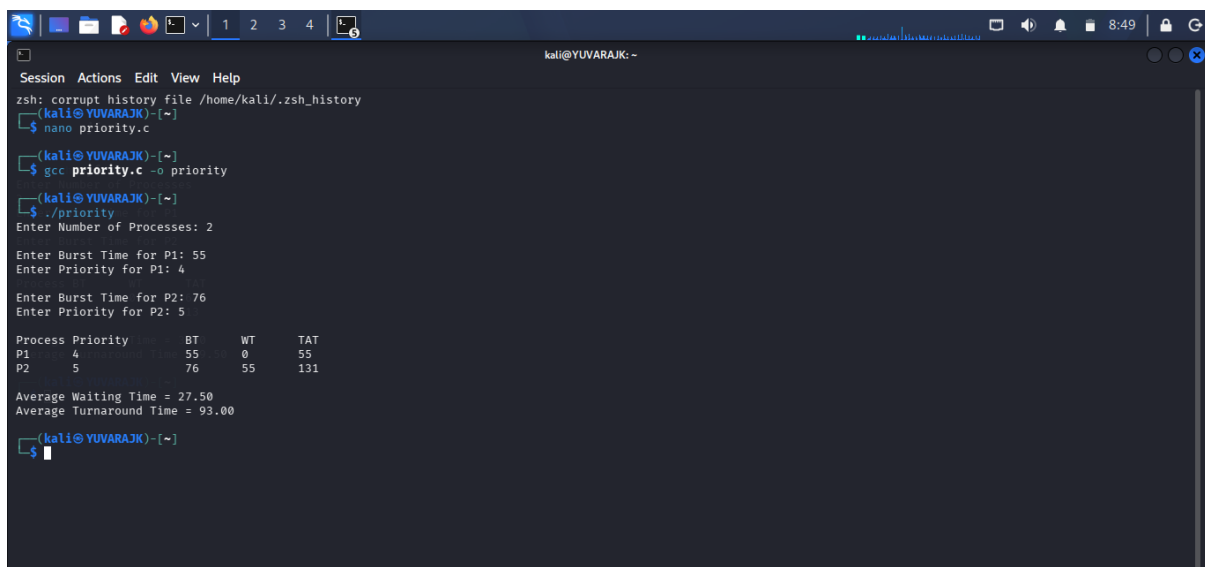
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano sjf.sh
(kali@YUVARAJK)-[~]
$ bash sjf.sh
Enter Number of Processes
2
Enter Burst Time for P1
6
Enter Burst Time for P2
7
Process BT      WT      TAT
P1      6        0        6
P2      7        6       13
Average Waiting Time = 3.00
Average Turnaround Time = 9.50
(kali@YUVARAJK)-[~]
$
```

C PROGRAMMING : PRIORITY SCHEDULING



```
GNU nano 9.0 priority.c *
#include<stdio.h>
int main() {
    int n,i,j,temp, bt[20],pr[20],wt[20],tat[20];
    float avg_wt=0, avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("\nEnter Burst Time for P%d: ",i+1);scanf("%d",&bt[i]);
        printf("Enter Priority for P%d: ",i+1);scanf("%d",&pr[i]);
        for(i=0;i<n-1;i++){
            for(j=i+1;j<n;j++){
                if(pr[i] > pr[j]){
                    temp = pr[i];
                    pr[i] = pr[j];
                    pr[j] = temp;
                    temp = bt[i];
                    bt[i] = bt[j];
                    bt[j] = temp;}}wt[0] = 0;
                for(i=1;i<n;i++){
                    wt[i] = wt[i-1] + bt[i-1];
                    printf("\nProcess\tPriority\tBT\tWT\tTAT\n");
                    for(i=0;i<n;i++){
                        tat[i] = wt[i] + bt[i];
                        avg_wt += wt[i];avg_tat += tat[i];
                        printf("P%d\t%d\t%d\t%d\t%d\n",i+1, pr[i], bt[i], wt[i], tat[i]);
                    }
                    avg_wt /= n;avg_tat /= n;
                    printf("\nAverage Waiting Time = %.2f", avg_wt);
                    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
                }
                return 0;
            }
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano priority.c
(kali@YUVARAJK)-[~]
$ gcc priority.c -o priority
(kali@YUVARAJK)-[~]
$ ./priority
Enter Number of Processes: 2

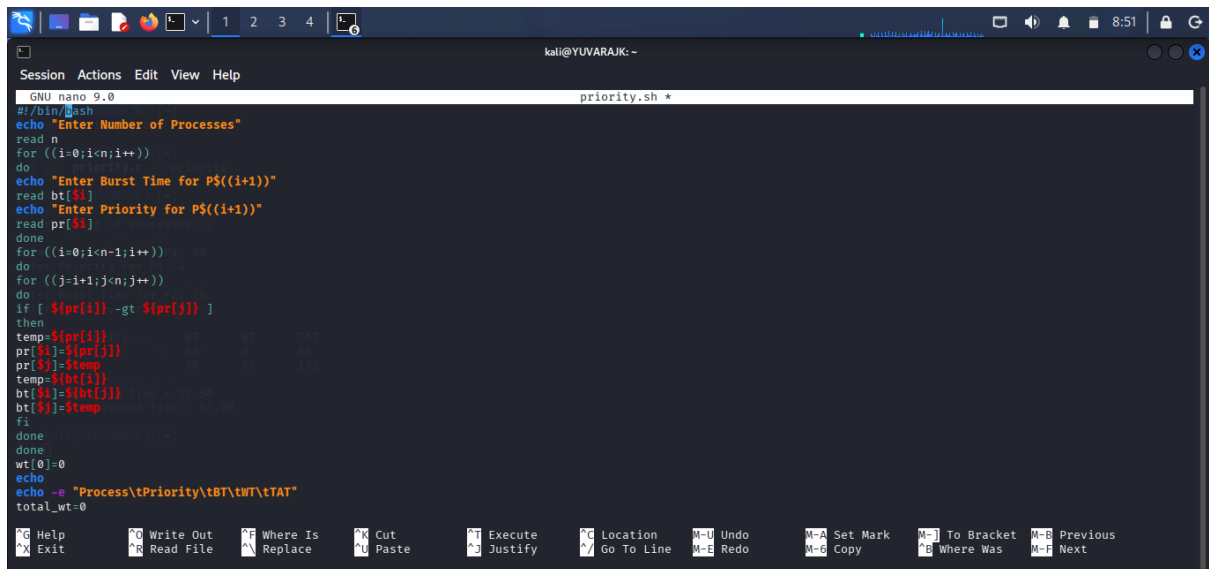
Enter Burst Time for P1: 55
Enter Priority for P1: 4

Enter Burst Time for P2: 76
Enter Priority for P2: 5

Process Priority      BT      WT      TAT
P1         4         55         0         55
P2         5         76         55        131

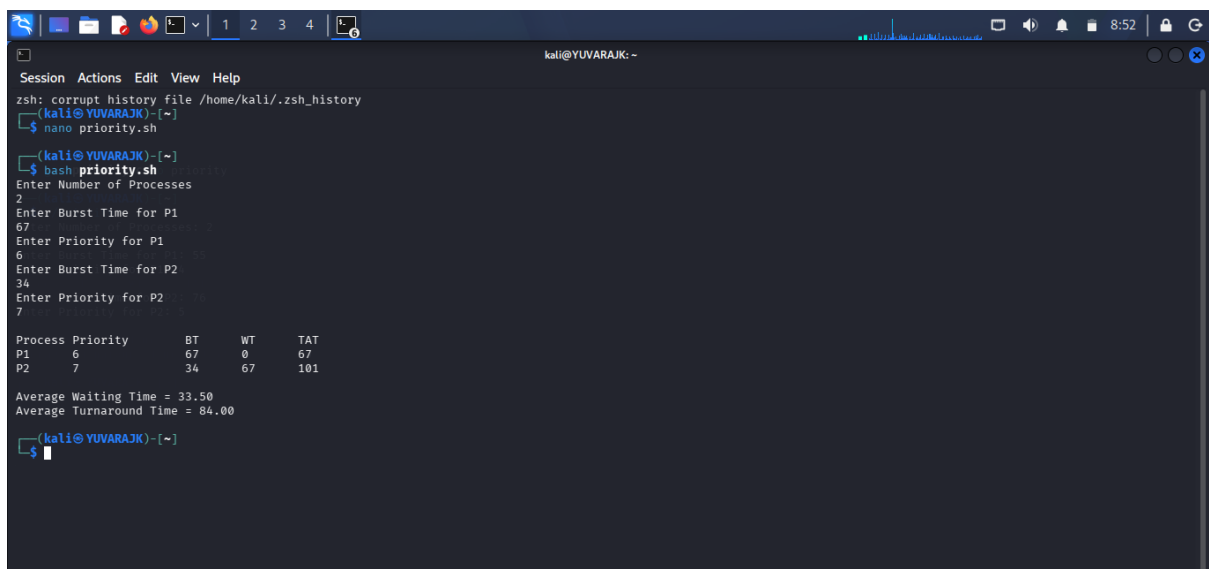
Average Waiting Time = 27.50
Average Turnaround Time = 93.00
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : SJF



```
GNU nano 9.0 priority.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P${(i+1)}"
read bt[$i]
echo "Enter Priority for P${(i+1)}"
read pr[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${pr[$i]} -gt ${pr[$j]} ]
then
temp=${pr[$i]}
pr[$i]=${pr[$j]}
pr[$j]=$temp
temp=${bt[$i]}
bt[$i]=${bt[$j]}
bt[$j]=$temp
fi
done
done
wt[0]=0
echo
echo -e "Process\tPriority\tBT\tWT\tTAT"
total_wt=0
for ((i=0;i<n;i++))
do
echo
done
```

OUTPUT :

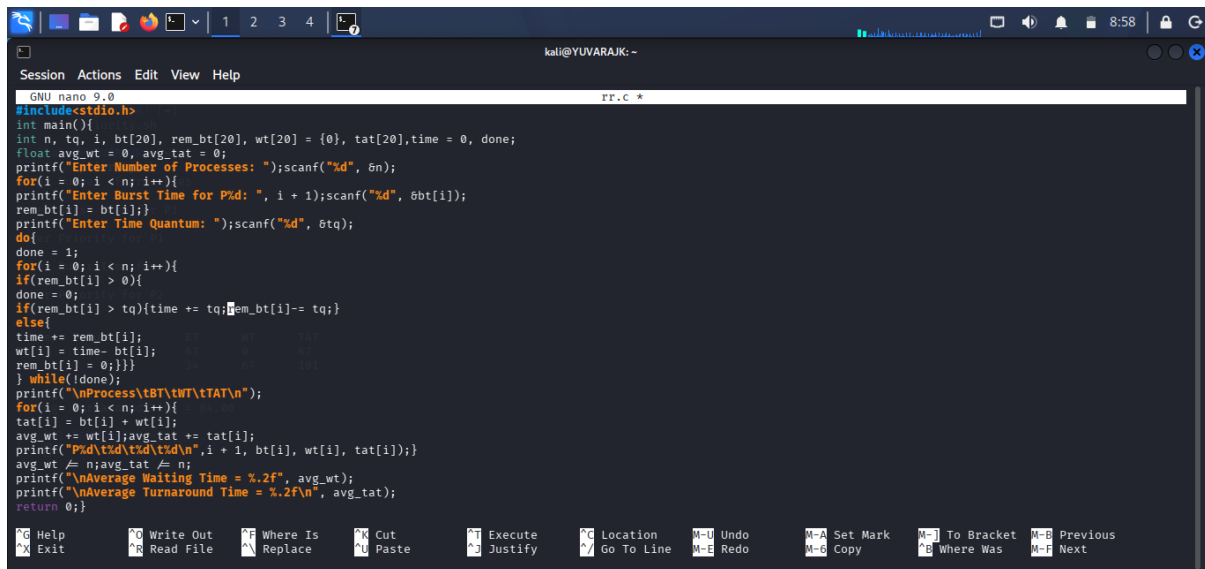


```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano priority.sh
(kali@YUVARAJK)-[~]
$ bash priority.sh
Enter Number of Processes
2
Enter Burst Time for P1
67
Enter Priority for P1
6
Enter Burst Time for P2
34
Enter Priority for P2
7

Process Priority      BT      WT      TAT
P1         6         67         67
P2         7         34        101

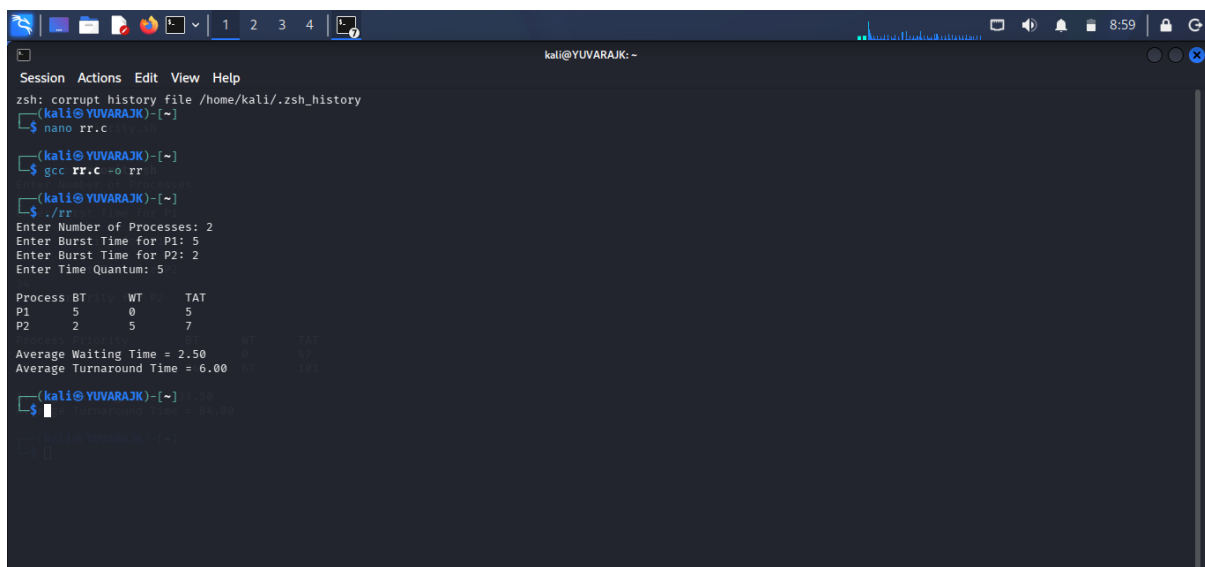
Average Waiting Time = 33.50
Average Turnaround Time = 84.00
(kali@YUVARAJK)-[~]
$
```

C PROGRAMMING : ROUND ROBIN



```
GNU nano 9.0 rr.c *
#include<stdio.h>
int main(){
int n, tq, i, bt[20], rem_bt[20], wt[20] = {0}, tat[20], time = 0, done;
float avg_wt = 0, avg_tat = 0;
printf("Enter Number of Processes: ");scanf("%d", &n);
for(i = 0; i < n; i++){
printf("Enter Burst Time for P%d: ", i + 1);scanf("%d", &bt[i]);
rem_bt[i] = bt[i];}
printf("Enter Time Quantum: ");scanf("%d", &tq);
do{
done = 1;
for(i = 0; i < n; i++){
if(rem_bt[i] > 0){
done = 0;
if(rem_bt[i] > tq){time += tq; rem_bt[i] -= tq;}
else{
time += rem_bt[i];
wt[i] = time - bt[i];
rem_bt[i] = 0;}}}
} while(!done);
printf("\nProcess\tBT\tWT\tTAT\n");
for(i = 0; i < n; i++){
tat[i] = bt[i] + wt[i];
avg_wt += wt[i];avg_tat += tat[i];
printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);}
avg_wt /= n;avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;}
```

OUTPUT :

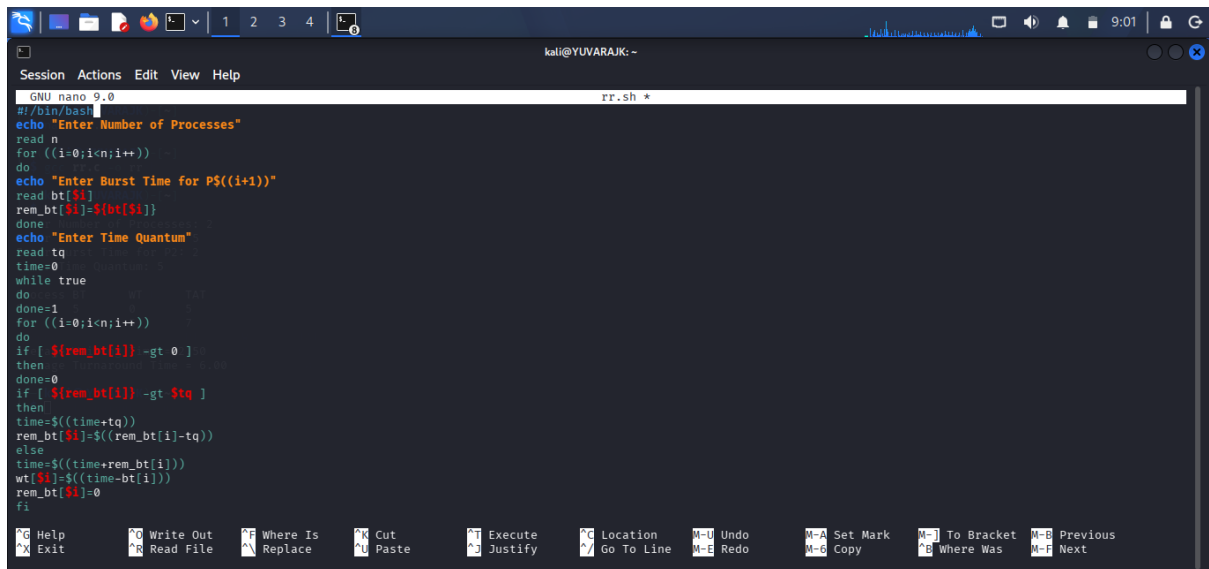


```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano rr.c
(kali@YUVARAJK)-[~]
$ gcc rr.c -o rr
(kali@YUVARAJK)-[~]
$ ./rr
Enter Number of Processes: 2
Enter Burst Time for P1: 5
Enter Burst Time for P2: 2
Enter Time Quantum: 5

Process BT      WT      TAT
P1      5        0        5
P2      2        5        7

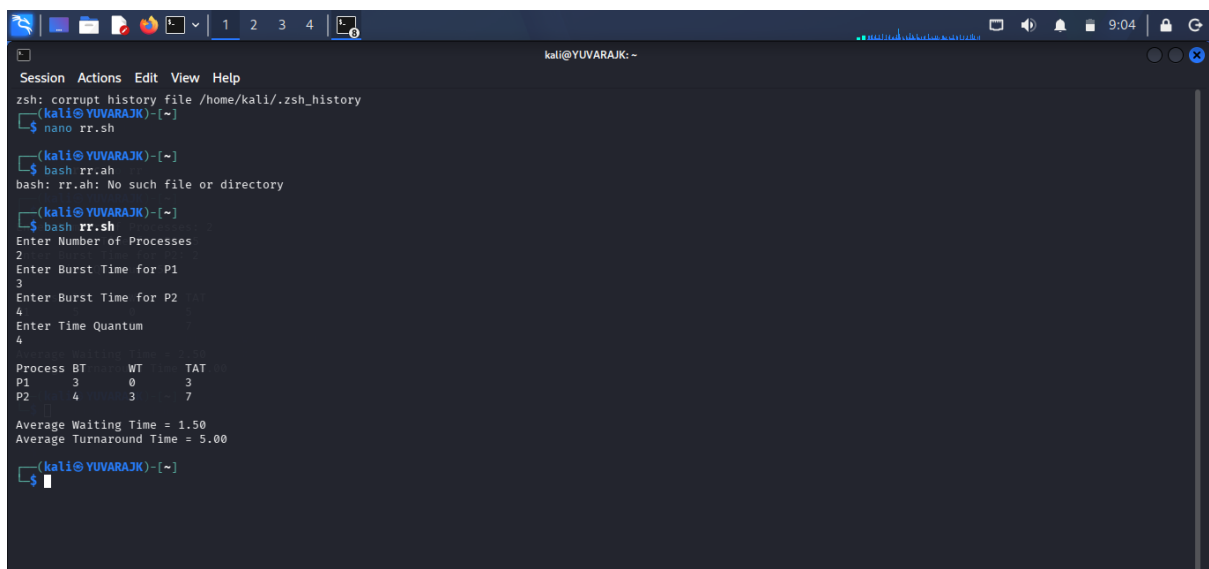
Average Waiting Time = 2.50
Average Turnaround Time = 6.00
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : ROUND ROBIN



```
GNU nano 9.0 rr.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P${(i+1)}"
read bt[$i]
rem_bt[$i]=$bt[$i]
done
echo "Enter Time Quantum"
read tq
time=0
while true
do
done=1
for ((i=0;i<n;i++))
do
if [ ${rem_bt[i]} -gt 0 ]
then
done=0
if [ ${rem_bt[i]} -gt $tq ]
then
time=$((time+tq))
rem_bt[i]=$((rem_bt[i]-tq))
else
time=$((time+rem_bt[i]))
wt[$i]=$((time-bt[i]))
rem_bt[i]=0
fi
fi
done
fi
fi
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@YUVARAJK: ~
$ nano rr.sh
kali@YUVARAJK: ~
$ bash rr.sh
bash: rr.sh: No such file or directory
kali@YUVARAJK: ~
$ bash rr.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
4
Enter Time Quantum
4

Process BT    WT    TAT
P1      3      0      3
P2      4      3      7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00
kali@YUVARAJK: ~
$
```